

Wazuh SIEM Home Lab: Attack Detection & Custom Rules

A hands-on SOC lab built with Wazuh. I deployed a SIEM, enrolled Windows and Linux endpoints, simulated common attacks in an isolated environment, and investigated the resulting alerts. I also wrote a custom detection rule and mapped the activity to MITRE ATT&CK.

All attacks were run only against my own machines inside a private lab network.

Lab Environment

Component	Details
SIEM server	Wazuh (manager, indexer, dashboard) on Ubuntu 25.04
Virtualization	Oracle VirtualBox, 6 GB RAM, bridged network adapter

Component	Details
Monitored endpoint 1	Windows 11 Pro with Wazuh agent 4.14.7 (agent name: Chandu-windows)
Monitored endpoint 2	The Ubuntu server itself (agent ID 000)
Attacker	The Windows host, using PowerShell and OpenSSH

Skills Demonstrated

SIEM administration, log analysis, alert triage, file integrity monitoring, custom rule writing (XML), MITRE ATT&CK mapping, Linux and Windows endpoint monitoring, troubleshooting.

Exercise 1: Windows Brute-Force Detection

Goal: Detect repeated failed logins on a Windows endpoint.

What I did: Locked the Windows session and entered a wrong password 6 times, then signed in correctly.

What Wazuh detected:

- **Rule 60122** (level 5), Logon Failure - Unknown user or bad password. Six alerts in about 17 seconds.
- **Rule 60118** (level 3), Windows Workstation Logon Success, when I signed in afterward.

Takeaway: The pattern of repeated failures followed by a success is a classic indicator of password guessing.

![Windows brute-force alert list]
(screenshots/windows-bruteforce-alert-list.png)

![Alert detail](screenshots/windows-bruteforce-details.png)

Exercise 2: File Integrity Monitoring (FIM)

Goal: Detect file creation, modification and deletion in a sensitive folder.

Configuration: Added a real-time monitored directory to the agent's `ossec.conf` (inside `<syscheck>`), then restarted the agent:

```
<directories realtime="yes"
  check_all="yes">C:\FIM_Test</directories>
```

What I did: Created, edited and deleted a `.txt` file in the folder. (The default ignore list skips extensions such as `.log` and `.png`, so I used `.txt`.)

What Wazuh detected:

Event	Rule	Level
File added	554	5
Integrity checksum changed (modified)	550	7
File deleted	553	7

Takeaway: Wazuh reports a file rename as a delete plus an add, which is useful to remember during investigations.

![FIM events](screenshots/fim_events.png)

Exercise 3: Custom Detection Rule

Goal: Raise the severity of new files created in a specific folder.

I added the following to

`/var/ossec/etc/rules/local_rules.xml` on the Wazuh server:

```
<group name="local,syscheck,">
  <rule id="100100" level="10">
    <if_sid>554</if_sid>
    <field name="file" type="pcre2">(?
i)c:\\fim_test</field>
    <description>Custom rule: new file
created in monitored FIM_Test
folder</description>
  </rule>
</group>
```

I validated the syntax with `wazuh-analysisd -t`, restarted the manager, and created a new file in the folder.

Result: Rule **100100** fired at **level 10**, while earlier events from the same folder had only produced the default level 5 alert (rule 554). The rule extends an existing rule (`if_sid`) and narrows it with a field match.

```
![Custom rule alerts](screenshots/custom-rule-
events.png)
```

Exercise 4: SSH Brute-Force Attack

Goal: Detect SSH password guessing against a Linux server.

What I did: Installed OpenSSH on the Ubuntu VM. From the Windows host, I repeatedly tried to log in with a non-existent user and with a real account using wrong passwords (10+ attempts).

What Wazuh detected (36 alerts in 15 minutes):

Rule	Level	Meaning
5710	5	Attempt to log in using a non-existent user
5760	5	sshd: authentication failed
5557 / 5503	5	Password check failed (unix_chkpwd, PAM)
2502	10	User missed the password more than one time

Investigation: Opening an alert showed the full picture: the attacker's source IP (`data.srcip` , my Windows host), the targeted account (`data.dstuser`), the raw log line (`full_log`) and the affected server (`agent.name: ubuntu`).

```
![SSH alerts](screenshots/ssh-alerts-list.png)
```

```
![SSH alerts, more detail](screenshots/ssh-alerts-list-2.png)
```

![Alert investigation](screenshots/ssh-alert-details.png)

Exercise 5: MITRE ATT&CK Mapping

Wazuh maps many of its rules to ATT&CK. Alerts from these tests mapped to:

Technique	Source of the alerts
T1110 / T1110.001, Brute Force / Password Guessing	SSH and Windows failed logons
T1021.004, Remote Services: SSH	SSH brute-force test
T1070.004, File Deletion	FIM deletion test

Tactics involved include Credential Access and Lateral Movement. Other techniques appeared in the dashboard from normal system activity (for example, `sudo` use), so I only listed the ones I could trace to a test I ran.

![MITRE ATT&CK techniques](screenshots/mitre-techniques.png)

![MITRE dashboard](screenshots/mitre-dashboard.png)

![MITRE dashboard, second view](screenshots/mitre-dashboard2.png)

![MITRE brute-force technique details](screenshots/mitre-bruteforce-technique-details.png)

![MITRE brute-force technique details, second view](screenshots/mitre-bruteforce-technique-details-2.png)

Troubleshooting Notes

- **Dashboard showed "Wazuh dashboard server is not ready yet".** The dashboard's stored indexer credentials no longer matched the indexer's password. Updating them in the dashboard keystore and restarting the service fixed it.
- **"Something went wrong: timeout of 20000ms exceeded".** I tested the manager API directly with `curl` (it answered in under a second, so credentials were correct), then restarted the dashboard after the API was up.
- **No FIM alerts at first.** Real-time monitoring only reports changes made after the agent restarts,

and the monitored folder must exist at that time. I confirmed this in the agent's `ossec.log`.

- **Time zones.** The dashboard shows UTC, which is 5 hours 30 minutes behind local time (India), so I compared alert times accordingly.

What I'd Do Next

- Add active response to block the attacking IP after repeated failures
- Ingest logs from a web server and detect web attacks
- Enable vulnerability detection and compliance (SCA) reporting
- Build a custom dashboard for failed-login trends

Repository Structure

```
.
├── README.md
├── screenshots/          # alert and
                           dashboard screenshots
├── rules/
    └── local_rules.xml   # custom rule
100100
```